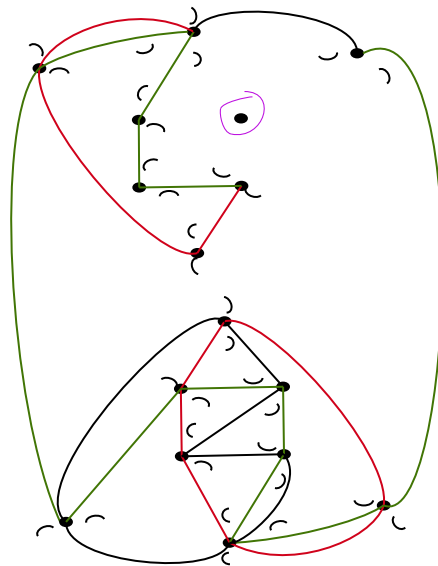
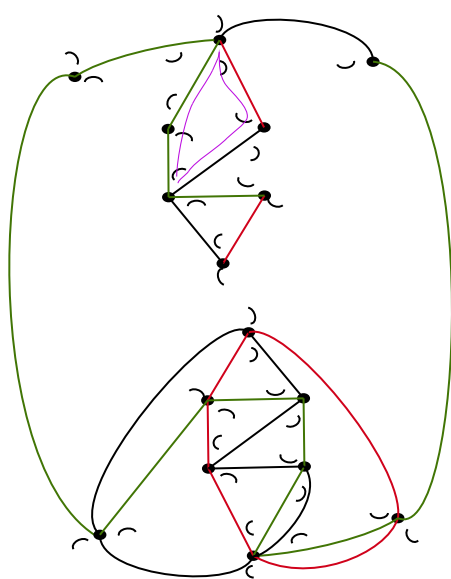
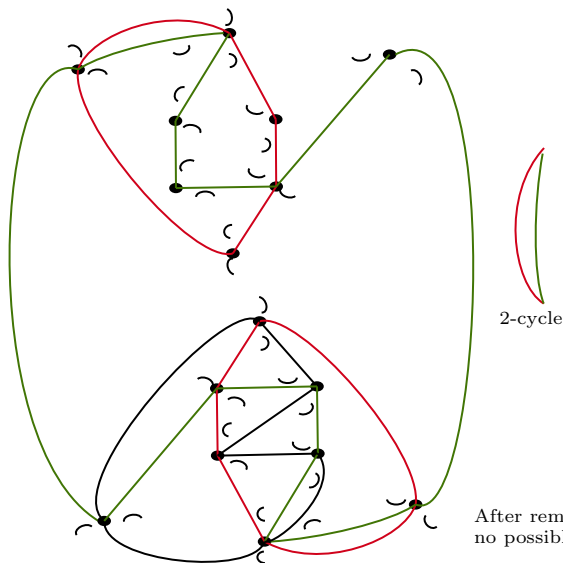


Disconnected $\backslash\mathit{math-container}$ after removing red cycle and then removing a compatible cycle using green and black edges.

Case with multiple compatible-cycle deletion:
 There are only two cycles of length 2 in $\backslash\mathit{math-container}$ and they don't produce a $\backslash\mathit{math-container}$, whether removed individually or with other vertex-disjoint compatible cycles.
 This compatible 2-cycle is also in $\backslash\mathit{math-container}$ but does not become a part of the cycle decomposition. There are no other compatible 2-cycles.
 Given all these observations there are no more possible compatible cycles to check.



Removal of longest possible compatible-cycles in one half of $\backslash\mathit{math-container}$ makes it disconnected or does not produce a

